**Semester I** **B. Tech CSE (AI)**

## BEE 101: INTRODUCTION TO ELECTRICAL AND ELECTRONICS ENGINEERING

**3L + 0T + 0P + 3 C** **MM l00**

**Unit 1: Basic Electrical Quantities: -**Basic concept of charge, current, voltage, resistance, power, energy and their units, Conversion of units of work, power and energy from one form to another. Electrical Energy: Difference between ac and dc and their applications, Classification of two terminal elements, Energy Sources, Source Transformation, ideal independent two terminal electrical sources.

**DC Networks**: Ohm's law, resistances, capacitances & inductors in series and parallel, Kirchhoff's laws and their applications in solving electrical network problems, Node Voltage and Mesh Current Analysis, Star-delta transformation.

**Unit 2: Network theorems:** Thevenin Theorem, Norton Theorem and Superposition Theorem, Statement and applications including dependent sources.

**Transformer:** Faraday's Law of Electromagnetic Induction, Construction and Operation of Single-Phase Transformer, EMF Equation, Voltage & Current Relationship and Phasor Diagram of Ideal Transformer.

**Unit 3: AC Fundamentals:** Concept of alternating current and voltage, Generation of Single-Phase AC Voltage, EMF Equation, Average, RMS and Effective Values Representation of alternating sinusoidal quantities by vectors, Phasor algebra (addition, subtraction, multiplication and division of complex quantities). RL, RC & RLC Series-Parallel Circuits, Complex Representation of Impedances, Reactance, Phasor Diagram, Power and Power Factor.

**Unit 4: Review of Semiconductors:** Semiconductors, conductors and insulators, intrinsic and extrinsic semiconductors and conduction in semiconductors**.**

**Semiconductor Physics:** Basic material properties of semi-conductors, governing factors for Fermi-level, carrier concentration and carrier mobility, recombination and carrier lifetime, carrier drift, diffusion, Hall effect and continuity equation.

**Unit 5: Junction Diodes:** Introduction, forward and reverse biasing of diode, voltage current characteristic of diode, diode models, half wave rectifier, full wave rectifier, Zener diode and its application.

**Bipolar Transistor:** Transistor structure, basic transistor operation, common base configuration and its characteristics, transistor amplifying action, common emitter configuration and its characteristics, common collector configuration, limit of operation, the dc operating point and biasing techniques (fixed bias, voltage divide bias, voltage feedback type, current feedback type, and combination of voltage and current feedback types)

**Text/Reference Books**


1. Integrated Electronics Analog and Digital Circuits & Systems, J. Millman & C.C. Halkias, McGraw Hill.
2 Electronic Devices and Integrated Circuits, B. P. Singh & Rekha Singh, Pearson Education.
3 Theory and problems of Basic Electrical Engineering, D.P. Kothari & I. J. Nagrath, PHI.
4 Basic Electrical Engineering, V.N. Mittal & A. Mittal, TMH.
5 Solutions of problems in Electrical Engineering, Smith Parker, CBS Publisher.
6 Electronics Devices & Circuit Theory, Boylestad & Nashelskey, Prentice Hall of India.